

Geostatistical Refinement of GIS Classification Results

Evaluating and Correcting Spatial Outliers via Kriging-based Hypothesis Testing

Weiqian Xu

Supervisor: Prof. Pankratova

Saint Petersburg State University
Applied Mathematics and Control Processes Department

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Presentation Outline

- 1 Introduction & Background
- 2 Exploratory Data Analysis
- 3 Methodology: Geostatistical Refinement Pipeline
- 4 Results and Analysis
- 5 Conclusion

Background: Pixel-based Classification in GIS

Context: Remote Sensing and Expectations Maximization (EM) Clustering

- High-resolution data demands automated, pixel-wise Land Use/Land Cover (LULC) classification or clustering.
- The EM clustering algorithm was used in the data used in this study, and five clustering results were given.

Tobler's 1st Law

"Everything is related to everything else, but near things are more related than distant things." [4]

The Weakness of Expectation-Maximization Algorithm: EM Clustering treat every single pixel as an independent sample. this directly violates Tobler's First Law of Geography. Natural landscapes are continuous and spatially autocorrelated.

Problem Statement: The 'Salt-and-Pepper' Effect

The Phenomenon: This happens when individual pixels, or small, fragmented groups, stand out against their background. Their labels, as assigned by the EM algorithm, clash with the surrounding data, creating a noisy, scattered appearance that doesn't match the environment. [1].

Consequences in GIS

- Reduced-quality LULC decision-making.
- Inaccurate spatial area estimation.
- Damaged landscape ecology metrics.

Limitations of Heuristic Filters

- **Majority Filter:** Purely geometric, removes thin linear features (e.g., roads, rivers) [3].
- **Markov Random Fields (MRF):** Tends to over-smooth real geographic boundaries.
- **Object-Based Image Analysis (OBIA):** Relies heavily on subjective parameters.

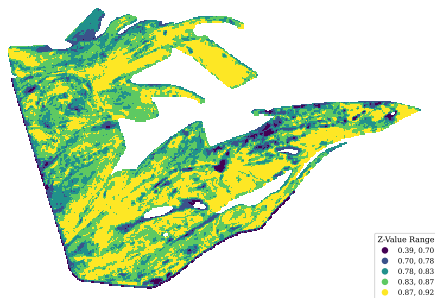
The Core Objective

Correct isolated labels so they make sense in the context of the surrounding physical landscape (Z).

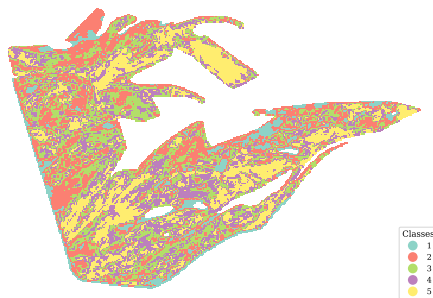
The Workflow:

- 1 **Identify isolated spots:** Use spatial connectivity metrics to flag pixels that stand out from their neighbors.
- 2 **Model the expected value:** Use Kriging to predict what the physical value should be at those locations.
- 3 **Test the consistency:** Check if the current labels actually match the underlying physical data.
- 4 **Reassign labels:** Apply Maximum Likelihood Estimation to reassign structurally inconsistent labels.

Data Overview: Spatial Distribution



(a) Productivity Distribution (Z)



(b) Fragmented Categorical Map (L)

Here, the continuous physical attribute Z is the productivity index, and the categorical map L is the result of EM clustering.

- The continuous physical field (a) shows attribute values range from 0.39 to 0.92, and shows a spatial gradient with clear spatial autocorrelation.
- The categorical map (b) shows high-frequency noise.

Verification of Prerequisite: Spatial Autocorrelation

Fundamental Assumption

Ordinary Kriging requires the attribute field $Z(\mathbf{X})$ to show spatial autocorrelation, rather than complete spatial randomness [2].

Quantitative Assessment (Global Moran's I)

$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij} (Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2} \quad (1)$$

- **Result:** A highly significant positive spatial autocorrelation ($I = 0.8002$, $p < 0.001$).
- **Implication:** This provides the theoretical support to model the physical field $Z(\mathbf{X})$ as a **Spatial Random Field (SRF)** and apply Kriging methods.

Normality Violation: Original data is apparently non-Normal distributed. This violates the normality assumption in later hypothesis testing, which will lead to biased predictions and invalid hypothesis testing [5].

Normal Score Transformation (NST)

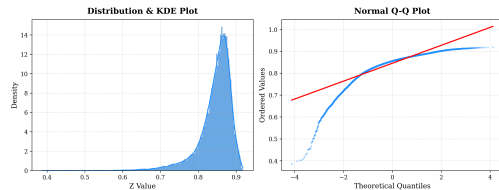
$$Y(\mathbf{X}_i) = \Phi^{-1}(F(Z(\mathbf{X}_i))) \quad (2)$$

* Φ^{-1} : Inverse standard normal CDF; F : Cumulative Distribution Function of Z .

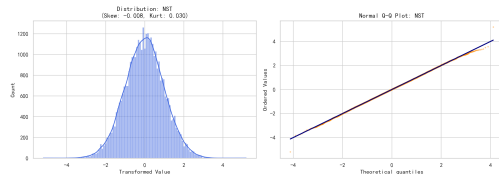
Transformation Results:

- Regularized spatial skewness to ≈ 0 .
- The empirical quantiles align with the Gaussian diagonal (see Q-Q plot).

Normality Assessment of Variable Z



(a) Original Z Histogram & Q-Q Plot



(b) NST Transformed Y Histogram & Q-Q Plot

Step 1: Isolation Detection (Candidate Extraction)

Objective: Identify spatially isolated pixels, which are likely misclassified rather than being true geographical features.

Mathematical Metrics

- **Neighbor Disagreement Score (NDS):** Measures spatial mismatch.

$$NDS(\mathbf{X}_i) = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbb{I}(L(\mathbf{X}_j) \neq L(\mathbf{X}_i)) \quad (3)$$

* $\mathbb{I}(\cdot)$ is the indicator function; $\mathcal{N}_i$ defines the Moore neighborhood.

- **Connected Components Analysis (CCA):** Evaluates spatial connectivity to flag clusters below the set Minimum Unit.

Algorithmic Logic: Denote $\mathcal{S}$ as the set of all pixels, $\mathcal{S}_{\text{NDS}}$ as the set of pixels with $NDS > \tau$ (e.g., $\tau = 0.8$), and $\mathcal{S}_{\text{CCA}}$ as the set of pixels in connected components smaller than the Minimum Unit (e.g., 4 pixels). The pixels within the union set $\mathcal{S}_{\text{cand}} = \mathcal{S}_{\text{NDS}} \cup \mathcal{S}_{\text{CCA}}$ are considered isolated and serve as prediction targets in the subsequent Kriging stage.

Step 2: Geostatistical Validation (Ordinary Kriging)

For candidate $\mathbf{X}_0 \in \mathcal{S}_{\text{cand}}$, we use Ordinary Kriging to predict the expected value $\hat{Z}(\mathbf{X}_0)$ and measure the associated uncertainty $\sigma_K^2(\mathbf{X}_0)$.

Kriging Methodology

$$\textbf{Estimator: } \hat{Z}(\mathbf{X}_0) = \sum_{i=1}^n \lambda_i Z(\mathbf{X}_i), \quad \mathbf{X}_i \in \mathcal{S}_{\text{reference}} = \mathcal{S} \setminus \mathcal{S}_{\text{cand}} \quad (4)$$

$$\textbf{Variance: } \sigma_K^2(\mathbf{X}_0) = 2 \sum_{i=1}^n \lambda_i \gamma(\mathbf{X}_i, \mathbf{X}_0) - \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j \gamma(\mathbf{X}_i, \mathbf{X}_j) \quad (5)$$

λ_i are the Kriging weights determined by solving the Kriging system of equations, γ is the semivariogram to model spatial dependence.

Why Ordinary Kriging?

- Best Linear Unbiased Predictor.
- Provides **physically grounded uncertainty** σ_K^2 .

Computational Optimization: The computation complexity is $\mathcal{O}(N^3)$, therefore uses only reference pixels that are within a certain distance from the candidate pixel.

Step 3: Hypothesis Testing Framework

- H_0 (**The "Normal" case**): The observed value at this location is consistent with the surrounding area. It looks like something that would naturally occur given its neighbors.
- H_A (**The "Outlier" case**): The observed value is out of place. It seems to come from a different process, meaning it doesn't fit in with the nearby environment.

Test Statistic $\eta(\mathbf{X}_0)$:

$$\eta(\mathbf{X}_0) = \frac{Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}(\mathbf{X}_0)}{\sigma(\mathbf{X}_0)} \sim \mathcal{N}(0, 1) \quad (6)$$

** Asymptotic normality is mathematically guaranteed by the prior NST step.*

To avoid the multiple testing problem across all candidates, we apply the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at a pre-specified level $q = 0.05$.

Decision Rule

- ➊ **Reject** H_0 : Large physical difference → **Keep original label** (Real feature).
- ➋ **Fail to reject** H_0 : Small difference → **Proceed to Reassignment** (Noise).

Step 4: Likelihood-Based Class Reassignment

Kriging-Based Likelihood Function

For each candidate class $C_k \in \mathcal{C}$, we compute the likelihood of observing $Z_{\text{obs}}(\mathbf{X}_0)$:

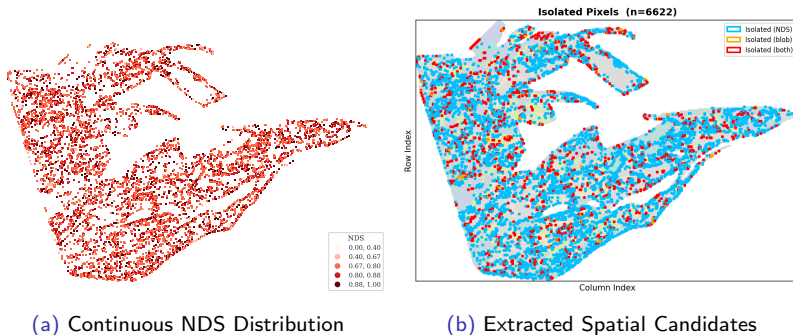
$$\mathcal{L}(C_k, \mathbf{X}_0) = \frac{1}{\sigma_{C_k}(\mathbf{X}_0)} \exp \left(-\frac{(Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}_{C_k}(\mathbf{X}_0))^2}{2\sigma_{C_k}^2(\mathbf{X}_0)} \right) \quad (7)$$

* $\hat{Z}_{C_k}(\mathbf{X}_0)$ and $\sigma_{C_k}^2(\mathbf{X}_0)$ are the Kriging prediction and variance computed using only the reference pixels belonging to class C_k , i.e., $\mathcal{S}_{\text{ref}}(C_k) = \{\mathbf{X}_i : L(\mathbf{X}_i) = C_k\}$.

- **Structural Regularization:** The leading term $\sigma_{C_k}^{-1}$ mathematically penalizes candidate classes that show high spatial uncertainty or limited local data.
- The corrected final label L^* is chosen to maximize the likelihood function across all candidate classes $\mathcal{C}$ (i.e., all 5 classes):

$$L^*(\mathbf{X}_0) = \arg \max_{C_k \in \mathcal{C}} \mathcal{L}(C_k, \mathbf{X}_0)$$

Result 1: Extraction of Candidate Outliers



Observations from Step 1:

- **NDS Map (a):** High spatial mismatches are primarily clustered along class boundaries.
- **Candidates (b):** Effectively isolates the "salt-and-pepper" noise patches. **6,622 pixels** flagged as candidates ($\approx 18.4\%$ of total domain).

Result 2: Semivariogram Modeling

Based on the weighted Root Mean Square Error (wRMSE), the Exponential model demonstrated the most optimal fit.

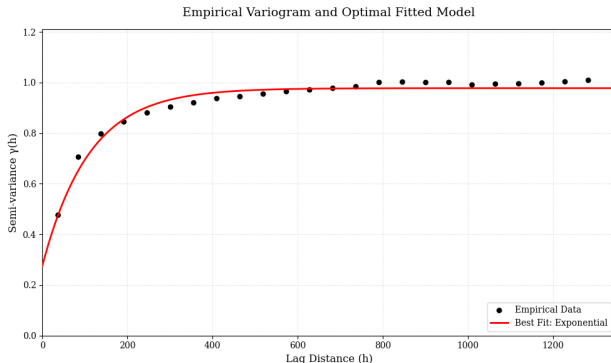
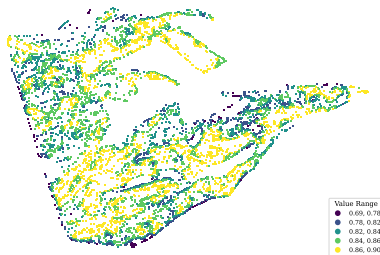


Figure 4: Empirical Semivariogram (black dots) and Fitted Models: Exponential (red).

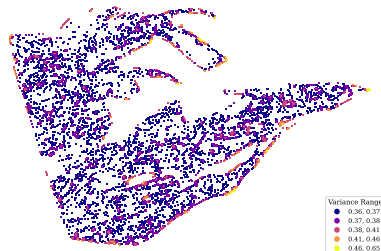
Fitted Parameters:

- **Nugget (c_0):** 0.275, reflects inherent sensor/environmental noise.
- **Sill (c):** 0.978, indicates the total variance of the spatial process.
- **Range (a):** 110.0 meters, defines the spatial correlation distance.

Result 3: Expected Realities and Uncertainty



(a) Kriging Prediction Field ($\hat{Z}$)



(b) Kriging Variance Field (σ_K^2)

- **Prediction (a):** Successfully maps the underlying smooth continuous field behind the categorical noise.
- **Variance (b):** Uncertainty is spatially measured ($\sigma_K^2 \in [0.36, 0.65]$). High variance zones correspond to areas with limited data support, guiding the subsequent hypothesis testing.

Result 4: Statistical Decisions

False Discovery Rate (FDR) Control at $q = 0.05$:

Statistical Category	Pixel Count	Percentage
Total Isolated Candidates (S_{cand})	6,622	100%
Rejected H_0 (Real Anomalies)	6	0.1%
Fail to reject H_0 (Artifacts/Noise)	6,616	99.9%

Interpretation & Impact:

- The visual "salt-and-pepper" noise is overwhelmingly confirmed as **non-physical artifacts** (99.9%).
- Crucially, the conservative FDR test successfully identifies and **keeps 6 real micro-structures** that would have been removed by standard filters.

Result 5: Label Correction Analysis (MLE)

MLE Reassignment Summary:

- **Reassigned (5,180):** Transitioned to a new label that maximizes the physical likelihood $\mathcal{L}(C_k, \mathbf{X}_0)$.
- **Retained (1,436):** Geometrically isolated, yet **statistically most probable** in their original class.

Key Insight

The Kriging-based MLE method effectively balances strong noise removal with the keeping of micro-scale geographic realities, resulting in a refined map that is both cleaner and more physically consistent.

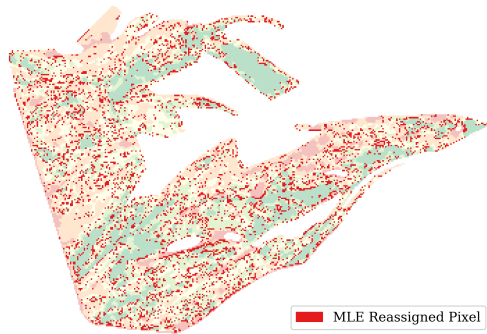
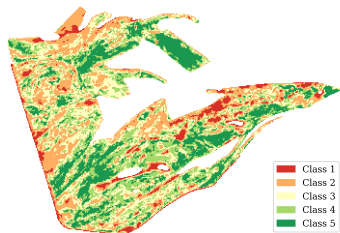
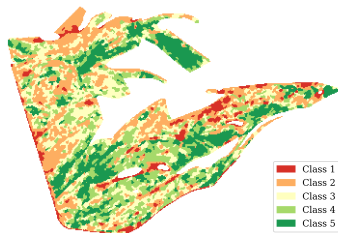


Figure 6: Final Refined Categorical Difference Map.

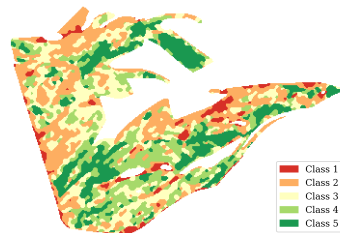
Comprehensive Map Comparison



(a) Original Fragmented Map



(b) Refined Map



(c) Majority Filtered Map

- The Kriging-based refinement (b) successfully removes the "salt-and-pepper" noise while keeping original micro-features, resulting in a more physically consistent map.
- In contrast, the Majority Filter (c) reassigns 9,542 pixels and over-smooths the map, removing thin linear features and creating artificial block-like patterns.

Research Summary:

- Transitioned post-classification from "geometric cleanup" to **formal spatial inference**.
- Successfully balanced strong noise removal with the keeping of micro-scale geographic realities.

Future Research Directions:

- Replace OK with **Universal Kriging** or **Regression Kriging** to include additional variables (e.g., DEM, Soil Moisture).
- Expanding $\gamma(\mathbf{h})$ into **3D semivariograms** $\gamma(\mathbf{h}, t)$ to validate pixels against their historical physical paths.
- Using cross-correlation between multiple spectral indices to improve refinement stability.

- [1] Julian Besag. “On the statistical analysis of dirty pictures”. In: *Journal of the Royal Statistical Society: Series B (Methodological)* 48.3 (1986), pp. 259–279.
- [2] Noel Cressie. *Statistics for spatial data*. John Wiley & Sons, 2015.
- [3] Kwang E Kim. “Adaptive majority filtering for contextual classification of remote sensing data”. In: *International Journal of Remote Sensing* 17.5 (1996), pp. 1083–1087.
- [4] Waldo R Tobler. “A computer movie simulating urban growth in the Detroit region”. In: *Economic geography* 46.2 (1970), pp. 234–240.
- [5] Hans Wackernagel. *Multivariate geostatistics: an introduction with applications*. Springer Science & Business Media, 2003.

Thank You!

Questions & Discussion

Weiqian Xu

Saint Petersburg State University

Applied Mathematics